

PrismQuant aligns activation geometry with the affine degrees of freedom of grouped asymmetric INT4. The fork starts from the same raw activations (a): Hadamard spreads energy (b), whereas PrismQuant aligns dominant eigendirections with quantizer-induced group-constant directions. Panels c and d are matched outcomes of these two rotations, not sequential transformations: they show the within-group max-minus-min ranges that determine quantizer scale, using identical token/group coordinates, camera, and 0–10 height/color normalization. Both branches use the same grouped asymmetric INT4 quantizer (group size 128). Panels c/d use one shared height colormap. The matrix, alignment, and quantizer glyphs are unscaled conceptual annotations, not additional measurements. The quantizer's affine offset represents a common group shift; subtracting an offset by itself does not change max-minus-min range.

The lower evidence strip retains the three measured signed activation traces: raw (e), Hadamard (f), and PrismQuant (g). Their range brackets are 11.656, 1.708, and 0.763, respectively, on a common signed-value scale. The dashed green line in g is the actual affine INT4 offset $\text{fp16}(\min x)$, not the group mean. These are trace measurements, not quantization-error or captured-energy plots.

All data are unchanged: Llama-3.2-3B, layer 27 down-projection inputs, non-BOS tokens 160–671 from sequence 118. Panels a/b retain channels 1254–3301 and independent magnitude scales 0–40 and 0–4. The channel window maximizes the number of channels with median $|x| > 1.0$ among all 6145 stride-1 windows of width 2048 (21 qualifying channels; earliest start breaks 183 tied maxima). Panels c/d retain every one of the 512×64 token/group ranges, including all peaks. Panels e/f/g show token 416 in raw group 25 and rotated group 0; f/g are exact cells of c/d. The layer and token were selected for illustration, not as an average across layers. Full selection rules and range statistics remain in `fig1_metadata.json`.